

Haoyuan Chen

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EDUCATION

Rensselaer Polytechnic Institute

Troy, NY

M.S. in Computer Science, Focus in Artificial Intelligence & Machine Learning

Jan. 2026 - Dec. 2026

*B.S. in Computer Science and Mathematics, Minor in Artificial Intelligence, **GPA: 3.94***

Aug. 2022 - Dec. 2025

Awards: Dean's Honors List, Archimedean Society, Rensselaer Leadership Award

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript/Typescript, HTML/CSS, SQL (PostgreSQL)

Frameworks/Libraries: ReactJS, Tailwind CSS, Redux, ExpressJS, NodeJS, NextJS, Prisma, PyTorch

Developer Tools: Git, VS Code, Eclipse, Microsoft Office, G-Suite, Postman API, Docker, Playwright, MongoDB

EXPERIENCE

Rensselaer Polytechnic Institute

Troy, NY

Undergraduate Teaching Assistant – Computer Science Department

Jan 2025 - Present

- Course Assistant for **Principles of Software** and **Intro to Artificial Intelligence** and Undergraduate Programming Mentor for **Computer Science I**.
- Facilitated **weekly lab periods and office hours**, along with other TAs to reinforce concepts taught in lectures.
- Graded **100+ students' homework and exams** while ensuring fair and consistent evaluation.

Harp Research Inc.

Troy, NY

Full-Stack Software Engineering Intern

May 2025 – Aug. 2025

- Collaborated in a team to create a **medical imaging workflow management system** that integrates with existing PACS (Picture Archiving and Communication System).
- Architected a **full-stack** system with a **Chrome extension** frontend, **Express** backend, and **PostgreSQL** database.
- Built **containerized microservices architecture** with **Docker Compose**, orchestrating multiple services including a dual-database setup (operational and demonstration databases).

PROJECTS

Privacy-Preserving Music Recommendation Engine | *PyTorch, Federated Learning, Differential Privacy, Streamlit*

- Built a **federated music recommender** on the Last.fm 1K dataset, transforming listening events into sparse user-artist interactions across 992 client datasets and training **Matrix Factorization and Neural Collaborative Filtering** models.
- Preserved user privacy by keeping user embeddings client-local, aggregating only shared model parameters, and applying gradient clipping with Gaussian noise.
- Evaluated held-out prediction and ranking quality with MSE, Hit@10, and NDCG@10; achieved a federated NCF test MSE of 1.6671 and deployed an interactive Streamlit demo that generates top-10 recommendations from artist embeddings without exposing raw listening histories.

Detecting Financial Fraud via GNNs | *PyTorch, PyTorch Geometric, scikit-learn, Pandas, Google Colab*

- Engineered a **reproducible fraud-detection pipeline** across PaySim, IEEE-CIS, and Elliptic datasets, including preprocessing, graph construction, class-imbalance handling, model training, threshold tuning, and evaluation.
- Developed and benchmarked **GCN, GAT, and GraphSAGE models** with full-batch and mini-batch training against Logistic Regression, MLP, and Random Forest baselines.

AI-Powered Personal Finance Analytics Platform | *Next.js, FastAPI, PostgreSQL, Redis, Docker, GitHub Actions*

- Built a full-stack asynchronous fintech system containerized via **multi-stage Docker configurations**, handling transaction ingestion streams.
- Implemented an automated **ML inference classification** engine leveraging text vectorization pipelines to sort unlabelled vendor strings with sub-millisecond inference latencies.
- Developed an interactive AI Copilot Agent utilizing OpenAI's function-calling capability to dynamically translate complex natural language text prompts into custom localized SQL database queries.
- Structured a predictive time-series spend forecast service using **Ridge L2 regularization models** to project upcoming monthly cashflow behavior while controlling for sparse cold-start volatility.